SUMMER INTERNSHIP PROGRAM · AI & ML · 2026

AI-Powered Fake News Detection

Classifying Real vs. Fake News Articles with Machine Learning & NLP

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Project-1 · Text Classification Pipeline for Real vs. Fake News

Why Do We Need Fake News Detection?

The scale of digital misinformation has outgrown manual review

1

Rapid Spread

Misinformation propagates across social media faster than fact-checkers can respond.

2

Impossible to Scale

Thousands of new articles are published daily — manual verification cannot keep pace.

3

Automated Defense

ML & NLP pipelines can flag deceptive content within milliseconds, at scale.

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The Goal

Build a system that reads an article and automatically classifies it as Real or Fake — with high accuracy and no human intervention.

44,898

news articles analyzed in this project

The Dataset

A balanced mix of real and fake news articles

44,898
Total Articles

23,481
Fake News (52%)

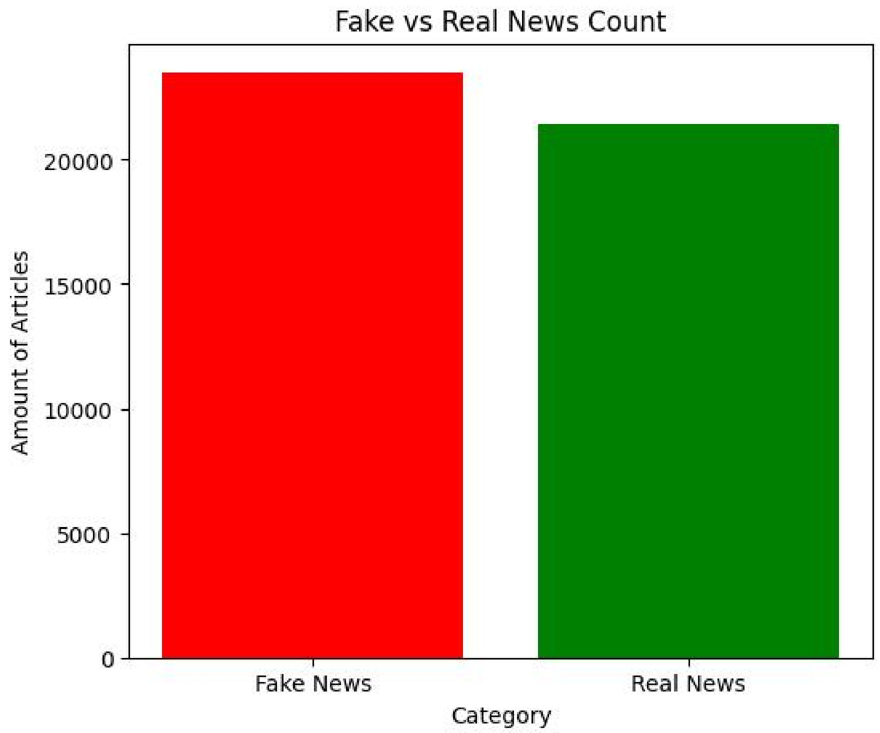
21,417
Real News (48%)

Feature used: the raw text content of each article (\`text\` column)

Target label: 0 = Fake News, 1 = Real News (engineered variable)

Split: 80% training (35,918) / 20% testing (8,980 articles)

Source datasets originally curated for fake news detection research, combining labeled real and fake article collections into a single corpus for supervised classification.



Class distribution: Fake vs. Real news counts

How Does the AI Learn?

Turning raw article text into a numerical signal a model can learn from

1

Lowercase

Normalize all text to lowercase vocabulary

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2

Remove URLs

Strip hyperlinks using regular expressions

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3

Strip Punctuation & Digits

Isolate pure semantic word content

>

4

TF-IDF Vectorize

Score word importance; top 5,000 features kept

Train / Test Split

80% Training · 35,918 articles	20% Test 8,980

Vocabulary constraint: TF-IDF was capped at the 5,000 most significant terms to balance model accuracy against computational cost.

Training Four Different Brains

Comparing parametric and non-parametric algorithms on the same TF-IDF features

K

K-Nearest Neighbors

Non-Parametric

Classifies an article by comparing it to its closest neighbors in feature space.

L

Logistic Regression

Parametric

Learns a linear decision boundary that separates fake from real vocabulary.

R

Random Forest

Ensemble · Non-Parametric

Combines hundreds of decision trees to vote on the final classification.

N

Neural Network (MLP)

Parametric

A multi-layer perceptron that learns nonlinear patterns in word usage.

Model Performance

Evaluated on the held-out 20% test set (8,980 articles)

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	**99.51%**	**99.62%**	**99.34%**	**99.48%**
Neural Network	98.75%	98.76%	98.62%	98.69%
Logistic Regression	98.33%	97.84%	98.67%	98.25%
K-Nearest Neighbors	**70.76%**	**91.68%**	**42.34%**	**57.93%**

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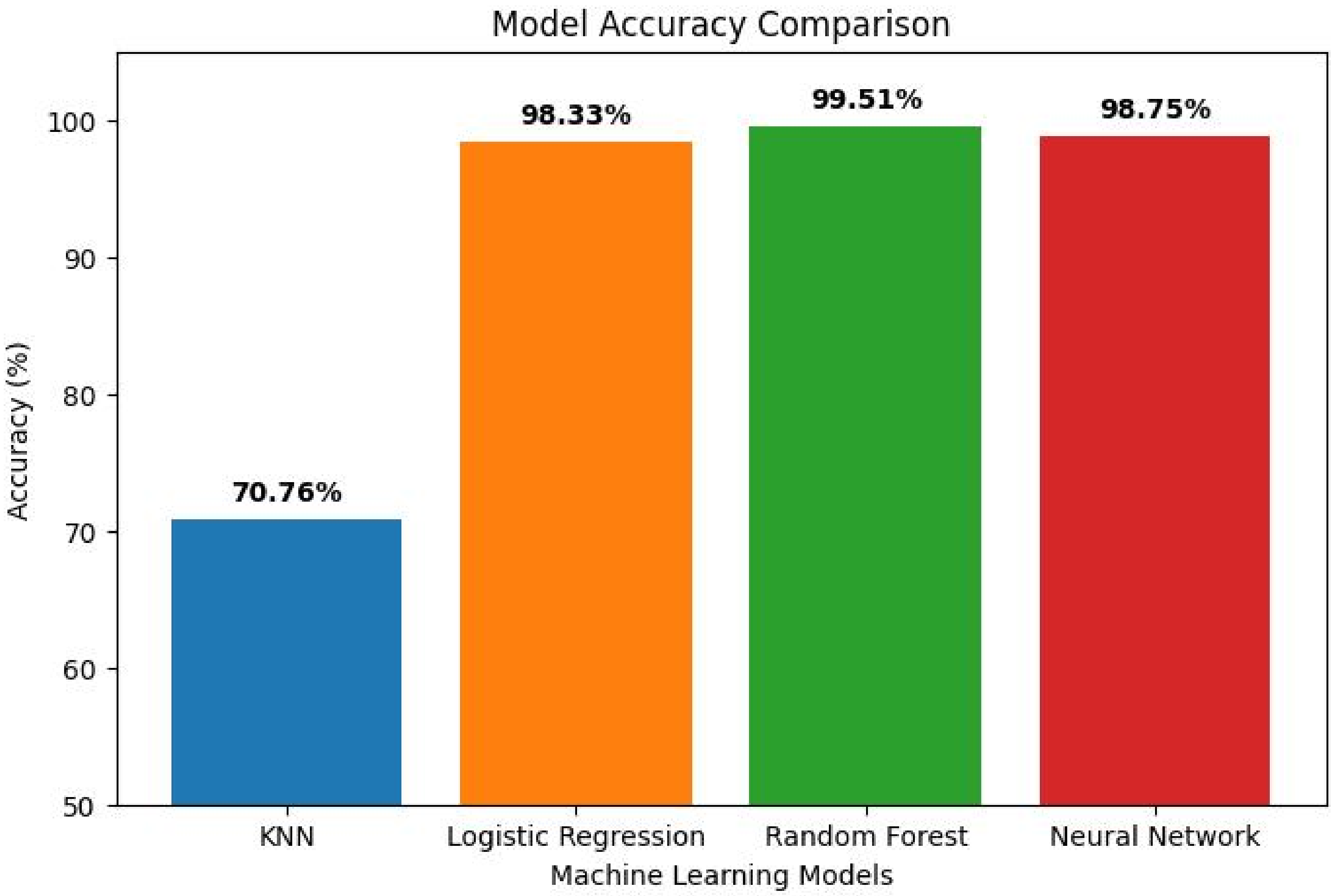
Weakest Model

KNN's recall dropped to just 42.3% — it missed over half of the fake articles in the test set.

Random Forest achieved the highest accuracy — roughly 44 misclassifications out of 8,980 test articles.

Accuracy at a Glance

Random Forest and the parametric models cluster near-perfect; KNN lags behind



What the Confusion Matrices Show

Random Forest

Roughly 44 total misclassifications out of 8,980 — near-perfect separation.

Neural Network & LogReg

About 112 and 150 total errors respectively, with strong balance across both classes.

KNN

Nearly 2,630 total misclassifications (29% error rate) — distance metrics struggle in 5,000-dim TF-IDF space.

Parametric vs. Non-Parametric Models

Two philosophies for learning from high-dimensional TF-IDF text data

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Parametric Models

Logistic Regression · Neural Network

Fix a set number of weights and assume a distribution over the data.

Both models exceeded 98% accuracy — evidence that the TF-IDF feature space is highly linearly separable.

Weight-based learning easily distinguishes fake vs. real vocabulary patterns.

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Non-Parametric Models

K-Nearest Neighbors · Random Forest

Make no fixed assumptions — model complexity adapts to the training data.

Random Forest (99.51%) was the best performer overall — its ensemble of trees handles sparse, high-dimensional data without overfitting.

KNN (70.76%) was the weakest — the curse of dimensionality cripples distance-based methods at 5,000 features.

Conclusion & Future Scope

ML and NLP can be powerfully leveraged to automate fake news detection — Random Forest delivered 99.51% accuracy, while Logistic Regression offered near-equal performance with far faster training.

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Limitations

- Relies on vocabulary patterns from a static, historical dataset
- May struggle with new slang or entirely novel events without retraining

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Future Scope

- Integrate real-time web scraping to evaluate live news articles
- Adopt transformer models (e.g. BERT) to capture context, not just word frequency

Thank you — Questions Welcome